

Transferable Structural Proposal Atlases for High-Dimensional Chance-Constrained Simulation Optimization

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Abstract

High-dimensional simulation optimization is statistically impossible under a tiny target budget without reusable structure. We study chance-constrained problems in which source tasks are available before a new target task is opened. We propose a dimension-equivariant structural proposal atlas: source outcomes rank normalized low-frequency policy profiles, target-label-free profiles provide universal support, and a maximin rule selects a small target initial design. The atlas is frozen before target outcomes and can be paired with any online optimizer. We prove that unconditional finite-budget coverage is impossible, give a conditional source-to-target coverage theorem whose geometry is independent of nominal dimension, and add an optimizer-independent terminal verifier with a familywise deployment-error bound. In three held-out synthetic domains at dimension 1,000, the atlas yields independently certified feasible recommendations in 60 of 60 trials with three different backends, whereas common Sobol initial designs yield at most one. At dimension 10,000 it certifies all 30 trials across budgets from 10 to 40 target search calls. An external electricity-storage holdout confirms an advantage over equal-total-cost unstructured search, but not over a natural low-frequency grid. The results identify transferable structural coverage, rather than a particular Bayesian-optimization backend, as the main source of sample efficiency.

Keywords: simulation optimization; chance constraints; transfer learning; high-dimensional Bayesian optimization; independent verification; structural priors

1 Introduction

Simulation optimization is often most valuable precisely where function evaluation is slow, noisy, and safety constrained. A policy may contain hundreds or thousands of coordinated

controls, while only a handful of target simulations are affordable. Examples include reserve schedules, multi-node inventory rules, and resource-allocation profiles. In these settings, the central statistical difficulty precedes the choice of an acquisition function: an unstructured initial design has negligible probability of intersecting a small feasible basin. A sophisticated sequential optimizer cannot improve a basin it never observes.

Historical tasks create a possible escape from this difficulty. Related systems may have been optimized in earlier markets, networks, or operating regimes. Their observations can reveal policy shapes that repeatedly balance objective quality and chance feasibility. Yet reusing those observations is delicate. Raw coordinates may change dimension, the meaning of individual controls may not align across tasks, and a target-specific hand-coded anchor would turn transfer into hidden oracle information. Moreover, an attractive recommendation and a deployment-safe recommendation are different statistical objects. Reusing the same observations for search and final certification can make the apparent evaluation budget and the safety claim both misleading.

This paper treats those issues as one information-design problem. Before a target task is opened, source outcomes are used to construct a small *structural proposal atlas*: a set of normalized policy profiles that is defined in a dimension-equivariant low-frequency coordinate. Source profiles are ranked by objective and chance-margin evidence, target-label-free profiles provide universal structural support, and a maximin rule fills a fixed initial design. The atlas is then frozen. Any target optimizer can consume it; in the primary implementation we use the standard sparse axis-aligned subspace Bayesian optimizer (SAASBO). After search, a separate verifier receives fresh samples from a frozen, ordered shortlist and never returns those samples to the optimizer.

The separation is substantive rather than cosmetic. It lets us ask three distinct questions. Does the transferred frontend reach a feasible basin? Does online optimization improve objective quality once coverage is established? Can the final recommendation be independently certified? It also forces a transparent cost accounting: source simulation, target search, and terminal verification are reported separately.

Our contributions are as follows.

- (i) We introduce a target-label-free, dimension-equivariant risk-objective proposal atlas. Its low-frequency coordinate has fixed dimension even when the raw policy dimension changes, and its source-monotone extension is fail-closed when source directions disagree.
- (ii) We establish the relevant theoretical boundary. No proper finite atlas can cover every possible held-out feasible set. Under an explicit source-to-target support condition, coordinate error bound, margin regularity, and safe-depth condition, however, the atlas contains a feasible policy. The sufficient condition depends on coordinate geometry

rather than nominal policy dimension. These and the terminal verification results are machine checked in Lean 4 without unproved placeholders.

- (iii) We design an optimizer-independent verifier with familywise control of unsafe deployment and an objective switch guard. Search, objective-comparison, and safety-verification samples are tracked separately, so a target search budget is never reported as a total simulation budget.
- (iv) We perform a causal frontend/backend evaluation. At dimension 1,000, the frozen atlas produces true-feasible and independently certified recommendations in all 60 trials with proposal-only, stacked-GP, and canonical SAASBO backends. Matched common-Sobol frontends produce zero, one, and zero, respectively. At dimension 10,000, all 30 trials are certified across target search budgets from 10 to 40 calls. An electricity-storage holdout confirms a large total-cost advantage over unstructured Sobol, while a post-confirmatory structured control shows no advantage over an obvious low-frequency grid.
- (v) We retain negative findings that delimit the contribution. Canonical SAASBO is a replaceable backend rather than a novel algorithm. A cumulative heteroscedastic variance model recovers variance shape much better than pooled variance but does not improve optimization outcomes, so it is reported as a calibration diagnostic rather than a second headline contribution.

The resulting claim is deliberately narrower than “solving arbitrary 10,000-dimensional optimization in ten evaluations.” The method succeeds when related tasks share a stable, low-dimensional structural coordinate; the paper states, audits, and stress-tests that condition. Section 2 positions the work. Sections 3–5 give the information contract, atlas, and guarantees. Sections 6 and 7 present the experiments, and Section 8 discusses limits and operational use.

2 Related Literature

2.1 Stochastic and Chance-Constrained Simulation Optimization

Simulation optimization allocates a finite evaluation budget among noisy alternatives when analytic objective and constraint functions are unavailable (Fu, 2002; Amaran et al., 2016). Stochastic kriging makes the simulation noise explicit and permits heteroscedastic metamodels (Ankenman et al., 2010). Constrained Bayesian optimization extends surrogate-guided search to noisy feasibility observations (Letham et al., 2019). Our departure is to make feasible-basin coverage a separate transferred design problem and to reserve fresh samples for terminal verification. The online optimizer can therefore change without changing the frontend or the deployment certificate.

2.2 High-Dimensional Bayesian Optimization

High-dimensional Bayesian optimization relies on structural restrictions. TuRBO localizes the search in adaptive trust regions (Eriksson et al., 2019); SCBO combines trust regions with constrained posterior sampling (Eriksson and Poloczek, 2021); and SAASBO imposes sparse axis-aligned length-scale priors (Eriksson and Jankowiak, 2021). These methods are strong target-only backends when enough target evaluations reveal relevant directions. They do not, by themselves, ensure that an initial design of ten points intersects a small chance-feasible basin in a thousand-dimensional integer cube. We use canonical SAASBO as the strongest audited backend, not as a methodological contribution, and compare against target-only TuRBO, SCBO, and SAASBO under both equal target budget and equal total simulation cost.

2.3 Transfer and Meta-Learning for Bayesian Optimization

Several lines of work transfer source information at different levels. FSBO learns a deep-kernel surrogate from source tasks (Wistuba and Grabocka, 2021). PACOH and Safe F-PACOH learn function-space priors with reliability guarantees (Rothfuss et al., 2021, 2023). HyperBO pretrains Gaussian-process priors from historical functions (Wang et al., 2024). Transfer GP methods model source-target discrepancy, while RGPE combines source experts according to target evidence (Tighineanu et al., 2022; Feurer et al., 2018). MetaBO and MALIBO transfer query utilities or acquisition behavior rather than only the surrogate (Volpp et al., 2020; Pan et al., 2024).

Our learning target is different. We transfer a finite *proposal measure* over normalized policy shapes before target observations, then expose the same initial design to multiple backend families. This makes the frontend’s causal contribution observable. It also provides a fair archive-level comparison: all transfer methods receive the same 384-call source archive, the same target budget, and the same independent terminal verifier.

2.4 Structural Priors, Coverage, and Certification

No-free-lunch results imply that optimization gains require assumptions about the function class or the distribution of tasks (Wolpert and Macready, 1997). Our theory makes that point concrete for finite target-label-free proposal sets and then states a positive geometric coverage condition. The dimension-equivariant coordinate is not claimed to be a universal representation; it is a low-frequency task hypothesis that can be audited and falsified.

Final safety introduces a second distinction. Posterior feasibility used during search is

model dependent, whereas deployment certification should be auditable independently of the acquisition rule. We use candidate-wise fresh-sample bounds and familywise error spending. For the external electricity experiment, an exact binomial bridge connects the all-success rule to the classical one-sided Clopper–Pearson lower confidence bound (Clopper and Pearson, 1934). This verification layer is deliberately more conservative and more expensive than a posterior label; its cost is reported rather than absorbed into the search budget.

3 Problem and Information Contract

3.1 Held-Out Chance-Constrained Task

Let q index a simulation task and let $\mathcal{X}_{q,d} = \{0, 1, \dots, L_q\}^d$ be its finite integer policy space. A simulation at $\mathbf{x} \in \mathcal{X}_{q,d}$ returns a noisy objective $F_q(\mathbf{x})$ and a noisy constraint response $G_q(\mathbf{x})$. The target task is

$$\min_{\mathbf{x} \in \mathcal{X}_{q,d}} f_q(\mathbf{x}) := \mathbb{E}[F_q(\mathbf{x})] \quad \text{subject to} \quad \mathbb{P}\{G_q(\mathbf{x}) \leq 0\} \geq 1 - \alpha. \quad (1)$$

For the Gaussian synthetic environments, the equivalent population chance margin is

$$m_q(\mathbf{x}) = \mu_{G,q}(\mathbf{x}) + z_{1-\alpha}\sigma_{G,q}(\mathbf{x}), \quad \mathbf{x} \text{ is feasible iff } m_q(\mathbf{x}) \leq 0, \quad (2)$$

where $z_{1-\alpha}$ is the standard normal quantile. The experiments use $\alpha = 0.05$. Two simulated objective outputs are scalarized with fixed weights $(1/2, 1/2)$; neither the weights nor the chance threshold is tuned by target seed.

The known synthetic oracle is used only after each run to score true feasibility and feasible regret,

$$r_q(\hat{\mathbf{x}}) = f_q(\hat{\mathbf{x}}) - \min_{\mathbf{x}: m_q(\mathbf{x}) \leq 0} f_q(\mathbf{x}), \quad (3)$$

and never for proposal construction, target search, or verification. Regret is undefined when the recommendation is truly infeasible, so feasibility counts and conditional regret are always reported together.

3.2 Source, Search, and Verification Information

Suppose Q_s source tasks have been completed before target task q is opened. Their archive is

$$\mathcal{D}_{\text{src}} = \{(q', \mathbf{x}_{q'i}, Y_{q'ir}^F, Y_{q'ir}^G) : q' \neq q\}, \quad (4)$$

with observed simulation outputs but no source optimizer, source feasible oracle, or target label. The final leave-one-domain-out protocol uses two source domains, 64 shared normalized profiles per source domain, and three independent replications per profile, for $S = 384$ source simulator calls.

Before observing a target response, a deterministic training map produces an atlas of at most n_0 target policies,

$$\mathcal{A}_{n_0} = T(\mathcal{D}_{\text{src}}, d, L_q, \text{schema}_q), \quad |\mathcal{A}_{n_0}| \leq n_0. \quad (5)$$

The target schema may identify dimension, integer bounds, and unlabeled policy or state channels. It may not contain target objective values, constraint values, feasible labels, target optimizers, or terminal verification outcomes. The atlas is immutable once the target stage begins.

The target optimizer first evaluates n_0 atlas points and may then request $N - n_0$ adaptive target evaluations. In the primary experiment $n_0 = 10$ and $N = 13$. After search, it freezes an ordered shortlist O_N and terminates. An independent verifier draws V additional target samples. These samples do not update the optimizer or the shortlist. Thus the exact simulation cost is

$$C_{\text{total}} = S + N + V, \quad (6)$$

and not N . We report source calls, target-search calls, objective-comparison calls, safety-verification calls, and total calls as separate fields.

3.3 Three-Part Decision Protocol

The protocol has three interfaces.

1. The *frontend* maps source observations and target descriptors to the frozen atlas $\mathcal{A}_{n_0}$.
2. The *backend* maps atlas and adaptive target observations to a shortlist. It may be proposal-only, a transfer GP, SC-V69, or canonical SAASBO.
3. The *verifier* maps the frozen shortlist and fresh samples to a deployment recommendation or no certificate.

This factorization is central to both theory and experiment. Atlas coverage does not depend on the backend. The verifier’s error bound does not depend on either the atlas or the backend. Conversely, success under a strong initial design cannot automatically be attributed to the online acquisition rule.

4 Dimension-Equivariant Structural Proposal Atlas

4.1 Low-Frequency Policy Coordinate

Write $z_i = x_i/L_q \in [0, 1]$ for the normalized policy. The raw policy length changes across tasks, so direct coordinate-wise transfer is not well defined. For a fixed frequency cap K , define

$$\bar{z} = d^{-1} \sum_{i=0}^{d-1} z_i, \quad s_z = \left[d^{-1} \sum_{i=0}^{d-1} (z_i - \bar{z})^2 \right]^{1/2}, \quad (7)$$

$$c_k(z) = \frac{2}{d} \sum_{i=0}^{d-1} (z_i - \bar{z}) \cos \left(\frac{\pi k(i + 1/2)}{d} \right), \quad k = 1, \dots, K. \quad (8)$$

The implemented coordinate $\hat{\eta}_d(\mathbf{x})$ contains $\bar{z}$, s_z , and a frequency-penalized selection of the coefficients in (8), followed by a fixed diagonal quadratic expansion. We use $K \leq 8$. Its dimension depends on K , not on d . The cosine basis also admits an inverse map: a coordinate is reconstructed as a constant profile plus the selected modes, clipped to $[0, 1]$, and rounded to the target integer bounds.

This coordinate encodes a specific structural hypothesis: slowly varying policy profiles and their coarse shape transfer more reliably than the identity of a raw coordinate. It does not assert that every high-dimensional problem is low frequency. Domain- and dimension-holdout experiments test whether this hypothesis remains useful.

4.2 Universal Support and Source Templates

The candidate library is the union of two components. The *universal structural support* contains target-label-free constant, monotone, and low-frequency profiles generated from (d, L_q) alone. It supplies coverage when source outcomes are misleading or do not include a useful endpoint. The *source templates* are normalized profiles observed in every source task.

For each source task, replicated observations estimate an objective score and a chance-margin score. Scores are converted to within-task percentiles before aggregation, avoiding a common numerical scale assumption across domains. Profiles that are robustly source feasible are preferred. Candidate templates on the source objective–risk Pareto frontier are selected under five fixed scalarizations: risk-only, objective-only, and three intermediate weights. This ranking uses observed source data only. It never uses a source analytic optimizer or a target outcome.

After the extreme Pareto templates are included, the remaining slots are filled by a

maximin rule in $\hat{\eta}$: at each step we select the candidate whose minimum coordinate distance to the current atlas is largest, with fixed penalties for source risk and objective rank. The result is a deterministic risk-objective atlas of size at most n_0 .

4.3 Fail-Closed Source-Monotone Envelope

A final source-only extension tests whether the zero-frequency coefficient $\bar{z}$ has a common chance-margin direction. For each source task q' , let $\rho_{q'}$ be the rank correlation between $\bar{z}$ and observed chance margin. With threshold $\kappa = 0.25$, the upper constant endpoint is admitted only if at least two source tasks all satisfy $\rho_{q'} \leq -\kappa$; the lower endpoint is admitted only if all satisfy $\rho_{q'} \geq \kappa$. Admission replaces an existing slot, so the budget remains n_0 . If neither condition holds, the update is the identity.

The original three synthetic domains reject this extension in all 60 primary cells, making the final version exactly equal to the base atlas there. The electricity family admits it using source markets before the confirmatory market is opened. Source agreement is evidence for a transferable direction, not proof that the target shares it; Section 5 states the additional target monotonicity condition explicitly.

4.4 Training and Deployment Algorithm

The full frontend is summarized below.

- A1.** Normalize the 64 shared profiles in each source task and compute replication-based objective and chance-margin summaries.
- A2.** Convert summaries to within-source percentile ranks; retain robust source-feasible Pareto templates under the fixed risk-objective weights.
- A3.** Form the candidate union of source templates and universal structural support; map all candidates into $\hat{\eta}_d$.
- A4.** Seed the atlas with selected Pareto extremes and fill remaining slots by the penalized maximin rule.
- A5.** Apply the fail-closed source-monotone endpoint test without changing atlas size.
- A6.** Freeze the atlas before the first target simulation. Evaluate its n_0 policies, then pass the observations to the chosen online backend.

The proposal is learned offline but is not free: its 384 source simulations are included in archive-fair and total-cost comparisons. When the same archive is reused for many targets, this cost may be amortized operationally; our tables do not assume such amortization.

5 Coverage and Verification Guarantees

This section separates three claims that are often conflated: a finite proposal cannot cover arbitrary targets; it can cover targets satisfying a structural transfer condition; and a recommendation can be verified without assuming the search model is correct. Complete machine-checked statements are mapped to the implementation in Appendix B.

5.1 Why an Explicit Transfer Condition Is Necessary

Theorem 5.1 (Finite-atlas no-free-lunch). *Let $\mathcal{X}$ be finite and let $\mathcal{A} \subsetneq \mathcal{X}$ be any proper proposal atlas frozen before target labels. There exists a nonempty feasible set $\mathcal{F} \subseteq \mathcal{X}$ such that $\mathcal{A} \cap \mathcal{F} = \emptyset$.*

Proof. Choose $x_0 \in \mathcal{X} \setminus \mathcal{A}$ and set $\mathcal{F} = \{x_0\}$. Then $\mathcal{F}$ is nonempty and is missed by $\mathcal{A}$. $\square$

Theorem 5.1 is elementary but decisive. Neither source learning nor a nonlinear encoder can guarantee that ten target-label-free proposals hit an arbitrary feasible set. A positive result must identify what transfers.

5.2 Aligned Geometric Atlas Coverage

Let η^* be an ideal transferable coordinate and $\hat{\eta}$ the coordinate learned from source information. Let $\mathcal{S}$ be the finite source-support library and $\mathcal{A}_{n_0} \subseteq \mathcal{S}$ the selected atlas. Distances below are in the common coordinate space.

Assumption 5.2 (Atlas coverage). For every $s \in \mathcal{S}$, there is $a \in \mathcal{A}_{n_0}$ such that $\|\hat{\eta}(s) - \hat{\eta}(a)\| \leq r_{\text{cover}}$, with $|\mathcal{A}_{n_0}| \leq n_0$.

Assumption 5.3 (Coordinate approximation). For each relevant policy x , $\|\hat{\eta}(x) - \eta^*(x)\| \leq \epsilon_\eta$.

Assumption 5.4 (Source-target support). There exists a target-safe center x_c and a source-support policy $s_c \in \mathcal{S}$ such that $\|\eta^*(x_c) - \eta^*(s_c)\| \leq \Delta_{\text{task}}$.

Assumption 5.5 (One-sided margin regularity and safe depth). The target chance margin satisfies

$$m_q(x) \leq m_q(y) + L_m \|\hat{\eta}(x) - \hat{\eta}(y)\| \quad (9)$$

for relevant x, y , and $m_q(x_c) + \gamma \leq 0$ for some $\gamma > 0$.

Table 1: Post-run finite-library coverage audit. Slack is safe radius minus the audited atlas/support radius. These diagnostics were not used for target selection.

Domain	r_{cover}	Support shift	Safe radius	Slack
FactorShock	0.0502	0.0000	0.3277	0.2775
Inventory	0.0307	0.0000	0.1322	0.1015
Queue	0.0384	0.0019	0.1298	0.0895

Theorem 5.6 (Aligned finite-atlas coverage). *Under Assumptions 5.2–5.5, if*

$$L_m(r_{\text{cover}} + \Delta_{\text{task}} + 2\epsilon_\eta) \leq \gamma, \quad (10)$$

then at least one $a \in \mathcal{A}_{n_0}$ is feasible for the target task.

Proof. Assumptions 5.3 and 5.4, followed by the triangle inequality, place s_c within $\Delta_{\text{task}} + 2\epsilon_\eta$ of x_c in the learned coordinate. Assumption 5.2 supplies $a \in \mathcal{A}_{n_0}$ at an additional distance at most r_{cover} . By (9) and (10),

$$m_q(a) \leq m_q(x_c) + L_m(r_{\text{cover}} + \Delta_{\text{task}} + 2\epsilon_\eta) \leq -\gamma + \gamma = 0.$$

Thus a is feasible. □

Corollary 5.7 (Nominal-dimension independence). *If r_{cover} , Δ_{task} , ϵ_η , L_m , and γ remain controlled as d changes, the sufficient condition (10) does not depend on d .*

Corollary 5.7 is a conditional effective-dimension result, not an assertion that the constants are automatically stable. The finite-library post-run audit verifies positive slack in all three synthetic domains (Table 1) under registered library quantities. It does not certify a global Lipschitz constant for every policy, so the global theorem remains an explicit modeling condition.

5.3 Fail-Closed Monotone Extension

Let U and L denote the upper and lower normalized constant policies. The source-monotone test of Section 4 either replaces one atlas entry with an admitted endpoint or returns the base atlas unchanged.

Proposition 5.8 (Fail-closed identity and budget). *If neither source direction satisfies the admission threshold, the extension returns the base atlas exactly. In all cases its cardinality is at most n_0 .*

Proposition 5.9 (Transferred endpoint safety). *Suppose all source tasks admit U , the target margin is nonincreasing in the zero-frequency coordinate, and U is coordinate-wise maximal. If any target policy is feasible, then U is feasible. A symmetric statement holds for the admitted lower endpoint and a nondecreasing target margin.*

Proof. For a feasible witness x_s , maximality gives $\bar{z}(x_s) \leq \bar{z}(U)$. Target monotonicity gives $m_q(U) \leq m_q(x_s) \leq 0$. The lower-endpoint case reverses the order. $\square$

The source rank statistic is only the admission rule. Target monotonicity is the additional identifiability condition and is not inferred by logic alone.

5.4 Optimizer-Independent Terminal Verification

After search, the backend freezes an ordered shortlist of at most three policies: an objective challenger, the backend recommendation, and a safe-interior support policy. Candidate j is tested with fresh samples and error allocation δ_j . Let E_j be the event that candidate j is certified even though it is unsafe.

Theorem 5.10 (Ordered-shortlist safety). *If $\mathbb{P}(E_j) \leq \delta_j$ for every frozen candidate, then the probability that the first certified shortlist member is unsafe is at most $\sum_j \delta_j$.*

Proof. Unsafe deployment implies E_j for at least one candidate that could be selected. The union bound yields the result; no property of the optimizer is used. $\square$

The challenger can replace a certified incumbent only if a one-sided upper confidence bound on their paired objective difference is negative. Let δ_{obj} bound failure of that objective upper bound.

Corollary 5.11 (Joint deployment and switch bound). *If $\sum_j \delta_j \leq \delta_{\text{safe}}$, then the probability of unsafe deployment or an incorrect objective switch is at most $\delta_{\text{safe}} + \delta_{\text{obj}}$.*

The implementation spends familywise error 0.05 across the frozen shortlist and uses eight fresh paired objective replications per compared policy when an objective switch is attempted. Verification samples never update the search posterior.

For the external energy experiment, one verification replication is a fresh calendar window and success means satisfying the reliability requirement in that window. If $K \sim \text{Binomial}(n, p)$, the probability of n successes is p^n .

Proposition 5.12 (Exact all-success certificate). *For any infeasible reliability probability $p \leq p_{\text{req}}$,*

$$\mathbb{P}_p(K = n) = p^n \leq p_{\text{req}}^n. \quad (11)$$

Error spending over a frozen finite shortlist gives the corresponding familywise bound.

The registered first stage uses 80 independent windows. At the 0.95 required probability and registered confidence level, the one-sided Clopper–Pearson rule certifies only the all-success count; the implementation equivalence is regression tested and Proposition 5.12 is machine checked from the binomial law.

5.5 Optional Cumulative Heteroscedastic Calibration

Some simulation tasks provide observable local exposures $A(x)$ and shared shock exposures $N(x)$. The optional cumulative factor model is

$$v_C(x) = A(x)^\top \Lambda A(x) + N(x)^\top B N(x) + N(x)^\top \omega + v_0, \quad (12)$$

with nonnegative diagonal Λ , positive-semidefinite B , and projected nonnegative linear/floor terms. This decomposition recovers local, shared-shock, linear, and floor contributions and is used only in the HVD diagnostic. The primary atlas and verifier do not require it. The distinction matters because better variance fit is not the same as better optimization; Section 7.6 reports both outcomes.

6 Experimental Design

6.1 Evaluation Questions

The experiment matrix is organized around five preregistered questions.

- Q1. Does the frozen atlas, rather than the online backend, cause feasible- basin coverage?
- Q2. Are both universal support and source templates needed?
- Q3. How does the method compare with transfer methods receiving the same source archive and target budget, and with target-only methods receiving the same total search cost?
- Q4. Does performance persist as raw dimension and the ratio d/N increase?
- Q5. Does the structural coordinate transfer to a held-out real-data control problem, and what happens against an obvious structured control?

A sixth diagnostic asks whether cumulative factor-HVD improves variance fit or operational optimization. It was governed by a promotion rule requiring both calibration improvement and at least one operational gain without harm.

6.2 Synthetic Task Family

We use three chance-constrained policy families. Each policy has integer entries in $\{0, \dots, 100\}$, giving 101^d raw alternatives.

FactorShock.

The objective and constraint depend on coarse policy level, tail level, and dispersion. Constraint noise contains three local exposures (level gap, tail gap, and spread) and two correlated shared-shock exposures (oscillation and tail shock), with the exact cumulative form in (12).

Inventory.

The policy is divided into stock, reorder, and safety-stock blocks. The objective balances holding, backlog, reorder, and safety losses; the chance constraint represents service reliability. Local variance exposures are backlog, holding, and stockout, while demand level and demand volatility are shared shocks.

Queue.

The policy is divided into capacity, priority, and smoothing blocks. The objective balances waiting and resource costs, and the chance constraint defines a safe operating pocket. Local exposures are queue, wait, and utilization; shared exposures are arrival bursts and common load.

These are not arbitrary high-dimensional functions. They deliberately embed different operations mechanisms in normalized, low-dimensional policy summaries and heteroscedastic noise. Their role is to test whether a dimension-equivariant profile learned from two mechanisms transfers to the third. In each leave-one-domain-out fold, the other two domains form the source archive. Each target domain uses 20 independent simulation seeds (80–99) at $d = 1000$ and fixed registered geometry. Truth is evaluated only post-run. The extreme-dimension study uses ten seeds per domain at $d = 10000$.

6.3 External Electricity-Storage Holdout

The external study uses hourly load, forecast, and price series from Open Power System Data (Open Power System Data, 2020). A $d = 1000$ integer profile controls storage reserve and state-of-charge behavior. The simulator includes charge and discharge limits, round-trip efficiency, reserve floor, cycling cost, spill, and unserved-energy penalties. The chance constraint limits the fraction of unserved demand windows.

The source markets are Austria 2017 and Denmark DK1 2017. Denmark DK2 2018 is the development target. Great Britain 2018 is opened once as the untouched 20-seed confirmatory target. The source-monotone endpoint rule is decided before Great Britain

outcomes. Search, audit, and verification use separated calendar samples. We compare the frozen atlas at $N = 13$ with (i) a natural constant-policy low-frequency grid at $N = 13$, added as a post-confirmatory fairness audit, and (ii) common Sobol with $N = 397$, equal to 384 source plus 13 target-search calls.

6.4 Methods and Fairness Contracts

The frontend/backend factorial uses two initial designs: the frozen atlas and common Sobol. Each is paired with proposal-only selection, a stacked transfer GP, and canonical SAASBO. Canonical means a BoTorch `SaasFullyBayesianSingleTaskGP` refit at every adaptive iteration (Balandat et al., 2020; Eriksson and Jankowiak, 2021); periodic-capped SAAS appears only as a separately named target-only computational control.

Archive-fair transfer comparisons receive the same 384 source calls, frozen $n_0 = 10$ initial design, target seeds, $N = 13$ target-search calls, and terminal verifier. They include Safe F-PACOH, RGPE, stacked GP, transfer multi-task GP, FSBO, HyperBO, MetaBO, and MALIBO. Official code paths are used with documented compatibility adaptations to the scalar constrained interface. Target-only TuRBO, SCBO, and periodic-capped SAAS receive $N = 397$ target-search calls, equal to the final method’s source-plus-target search cost. All final comparisons use the same independent verifier.

The frontend component ablation retrains and regenerates every proposal from the same source archive. It compares universal support only, source templates only, and their combined maximin atlas. This is the supported structural ablation. Earlier orthogonality, sparsity, additive, manifold, transformer, KG, and Thompson-sampling variants remain development diagnostics and are not recast as coequal contributions.

6.5 Verification, Metrics, and Inference

The synthetic verifier uses an ordered three-policy shortlist with candidate budgets 80, 128, and 128, familywise error 0.05, and a fresh paired objective guard when switching incumbents. It may stop after the first candidate. Consequently, verification calls are variable and are reported as a cost, not normalized away. The energy verifier uses 80 fresh windows in its first stage and the exact binomial/Clopper–Pearson rule in Proposition 5.12.

We report true-feasible recommendations, independently certified recommendations, false certificates, feasible regret conditional on true feasibility, and verification cost. Post-run HVD diagnostics report log-variance RMSE and variance-shape correlation over a fixed audit set. The statistical plan uses paired seeds, 10,000-sample bootstrap intervals for median paired differences, exact McNemar tests for paired binary outcomes, Wilcoxon signed-rank tests for

Table 2: Frontend/backend factorial at $d = 1000$. Frozen-atlas arms use $n_0 = 10$ and at most three adaptive calls; proposal-only stops at ten. Every arm uses the same independent verifier.

Domain	Frontend	Backend	Feasible	Certified	False	Med. regret
FactorShock	Frozen atlas	Proposal only	20/20	20/20	0	0.0082
FactorShock	Frozen atlas	Stacked GP	20/20	20/20	0	0.0082
FactorShock	Frozen atlas	Canonical SAASBO	20/20	20/20	0	0.0082
FactorShock	Common Sobol	Proposal only	0/20	0/20	0	–
FactorShock	Common Sobol	Stacked GP	0/20	0/20	0	–
FactorShock	Common Sobol	Canonical SAASBO	0/20	0/20	0	–
Inventory	Frozen atlas	Proposal only	20/20	20/20	0	0.0116
Inventory	Frozen atlas	Stacked GP	20/20	20/20	0	0.0116
Inventory	Frozen atlas	Canonical SAASBO	20/20	20/20	0	0.0031
Inventory	Common Sobol	Proposal only	0/20	0/20	0	–
Inventory	Common Sobol	Stacked GP	1/20	1/20	0	0.0108
Inventory	Common Sobol	Canonical SAASBO	0/20	0/20	0	–
Queue	Frozen atlas	Proposal only	20/20	20/20	0	0.0029
Queue	Frozen atlas	Stacked GP	20/20	20/20	0	0.0029
Queue	Frozen atlas	Canonical SAASBO	20/20	20/20	0	0.0015
Queue	Common Sobol	Proposal only	0/20	0/20	0	–
Queue	Common Sobol	Stacked GP	0/20	0/20	0	–
Queue	Common Sobol	Canonical SAASBO	0/20	0/20	0	–

paired continuous outcomes, and Holm adjustment within preregistered confirmatory families. Domain strata are descriptive heterogeneity analyses, not separate confirmatory families. Failed and timed- out runs remain in the immutable registry; all release-required cells used below are complete.

7 Results

7.1 The Frontend Determines Feasible-Basin Coverage

Table 2 gives the main causal comparison. With the frozen atlas, all three backends produce true-feasible and independently certified recommendations in all 60 domain-seed pairs, with no false certificate. Replacing only the frontend by common Sobol reduces these counts to 0/60 for proposal-only, 1/60 for stacked GP, and 0/60 for canonical SAASBO. For the preregistered proposal-only and SAAS comparisons, the Holm-adjusted McNemar p -value is 6.94×10^{-18} for both feasibility and certification. Figure 1 shows the aggregate contrast.

Once coverage is established, the backend affects objective quality. Across all 60 frozen-atlas pairs, canonical SAASBO improves regret over proposal-only in 29 seeds, loses in

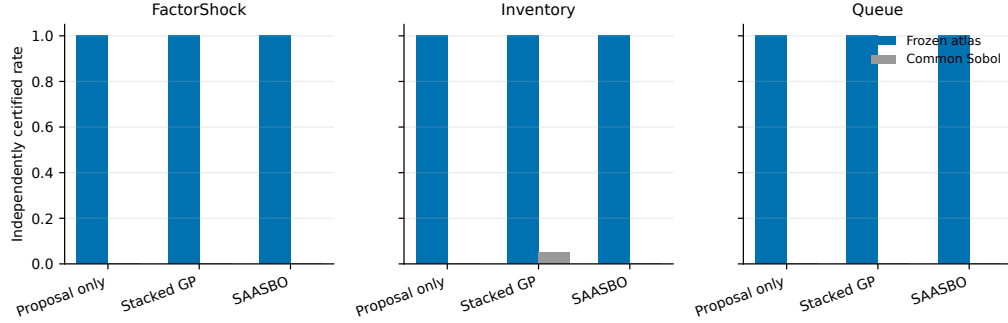


Figure 1: Feasible-basin coverage is driven by the frozen structural frontend, not by the backend. Counts aggregate 20 seeds in each of three domains.

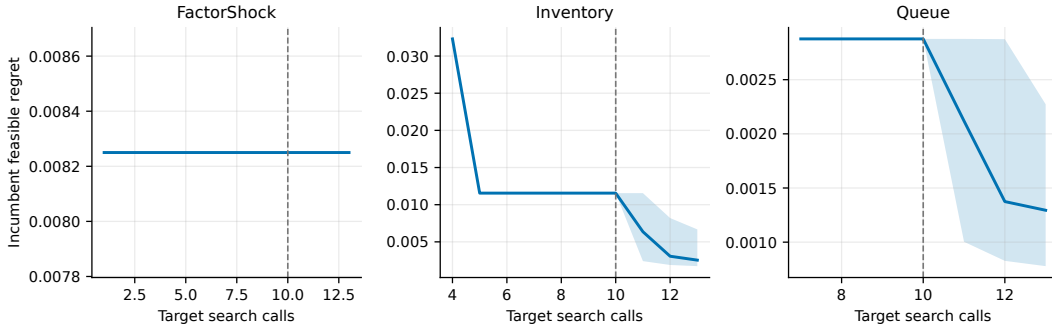


Figure 2: Median best true-feasible objective gap during target search. Missing early values mean that no evaluated point was truly feasible at that call. Verification samples are excluded.

two, and ties in 29; the Holm-adjusted Wilcoxon p -value is 1.16×10^{-6} . It improves over stacked GP in 29 seeds, loses in three, and ties in 28 ($p_{\text{Holm}} = 1.49 \times 10^{-6}$). The median paired difference is zero because FactorShock and many other pairs tie, while Inventory and Queue contain the improvements. The target-stage convergence paths in Figure 2 make the decomposition clear: the initial atlas reaches the feasible region, and the three adaptive calls occasionally improve its objective.

7.2 Both Atlas Components Matter

Universal structural support alone reaches 27/60 feasible trials; source templates alone reach 40/60; their combined maximin atlas reaches 60/60 (Table 3 and Figure 3). Universal support is particularly important as a source-misspecification fallback, whereas source templates concentrate probability mass around empirically safe, low-cost profiles. The combined atlas is therefore not merely the best source profile copied to a new dimension.

Table 3: Causal frontend component ablation with the same source archive, target seeds, proposal budget, and verifier.

Frontend component	Feasible	Certified	False	Med. regret
Universal support	27/60	23/60	0	0.0082
Source templates	40/60	40/60	0	0.0072
Combined maximin atlas	60/60	60/60	0	0.0082

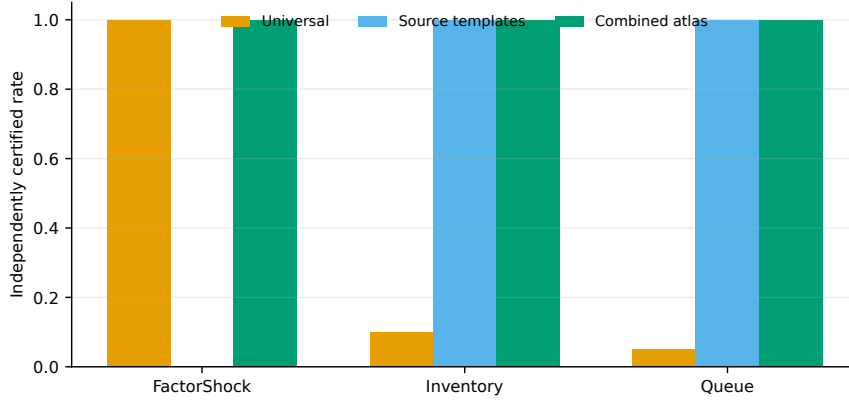


Figure 3: Feasible and independently certified recommendations under the frontend component ablation.

7.3 Archive-Fair Transfer and Total-Cost Controls

Table 4 compares transfer methods under exactly the same source archive, frozen initial design, target budget, seeds, and verifier. Because every method receives the strong atlas, most achieve high feasible coverage. Safe F-PACOH reaches 56/60, RGPE and FSBO 59/60, transfer MTGP 57/60, and stacked GP, HyperBO, MetaBO, MALIBO, and canonical SAASBO 60/60. Canonical SAASBO has the lowest aggregate median feasible regret (0.0038), whereas the other displayed methods have 0.0082. Verification cost differs materially, demonstrating why the 60/60 coverage count alone is insufficient.

The equal-total-search-cost comparison is more severe. Target-only TuRBO and SCBO receive 397 target-search calls and find no feasible recommendation in 60 trials. Periodic-capped target-only SAAS finds two, both certified. The atlas plus canonical SAASBO uses 384 source calls and 13 target-search calls and certifies all 60 (Table 5). The Holm-adjusted McNemar p -values for the final method versus each target-only control are below 3×10^{-17} . These results support reuse of related source simulations; they do not imply that source calls are free or that the frontend is uniformly better than every task-specific structured design.

Table 4: Archive-fair transfer comparison at $d = 1000$, $S = 384$, $n_0 = 10$, and $N = 13$. Mean verification includes safety and any fresh objective- comparison calls.

Method	Feasible	Certified	False	Med. regret	Mean verify
Atlas + canonical SAASBO	60/60	60/60	0	0.0038	173.9
Safe F-PACOH	56/60	56/60	0	0.0082	215.5
RGPE	59/60	59/60	0	0.0082	154.9
Stacked GP	60/60	60/60	0	0.0082	90.9
Transfer MTGP	57/60	57/60	0	0.0082	204.3
FSBO	59/60	59/60	0	0.0082	181.6
HyperBO	60/60	60/60	0	0.0082	211.7
MetaBO	60/60	60/60	0	0.0082	158.9
MALIBO	60/60	60/60	0	0.0082	162.9

Table 5: Equal source-plus-search-cost comparison. Verification is applied after search and reported separately. “Target-only SAASBO” denotes the audited periodic-capped control, not canonical every-iteration SAASBO.

Method	Source	Search	Feasible	Certified	False
Atlas + canonical SAASBO	384	13	60/60	60/60	0
Target-only TuRBO	0	397	0/60	0/60	0
Target-only SCBO	0	397	0/60	0/60	0
Target-only SAASBO	0	397	2/60	2/60	0

7.4 Dimension and Target-Budget Frontier

At $d = 1000$, proposal-only at ten search calls and canonical SAASBO at 13 both certify all 60 trials; aggregate median regret improves from 0.00825 to 0.00383. At $d = 10000$, all 30 trials are feasible and certified with zero false certificates at search budgets 10, 13, 20, and 40 (Table 6). The corresponding d/N ratios range from 1,000 to 250. Median regret is 0.00825, 0.00825, 0.00451, and 0.00579. The deterioration from $N = 20$ to $N = 40$ is retained rather than smoothed away; additional adaptive search is not monotonically beneficial under this finite stochastic protocol.

The result is consistent with Corollary 5.7: the raw dimension can grow while the profile coordinate remains fixed. It is not evidence that SAASBO or the atlas can identify an arbitrary sparse set of raw coordinates in ten calls. Indeed, the common-Sobol negative controls show that the same backend without structural coverage fails at $d = 1000$.

Table 6: Dimension and target-search budget. Source archive size remains 384; verification is additional and variable.

d	Search	d/N	Source	Feasible	Certified	False	Med. regret
1000	10	100.0	384	60/60	60/60	0	0.0082
1000	13	76.9	384	60/60	60/60	0	0.0038
10000	10	1000.0	384	30/30	30/30	0	0.0082
10000	13	769.2	384	30/30	30/30	0	0.0082
10000	20	500.0	384	30/30	30/30	0	0.0045
10000	40	250.0	384	30/30	30/30	0	0.0058

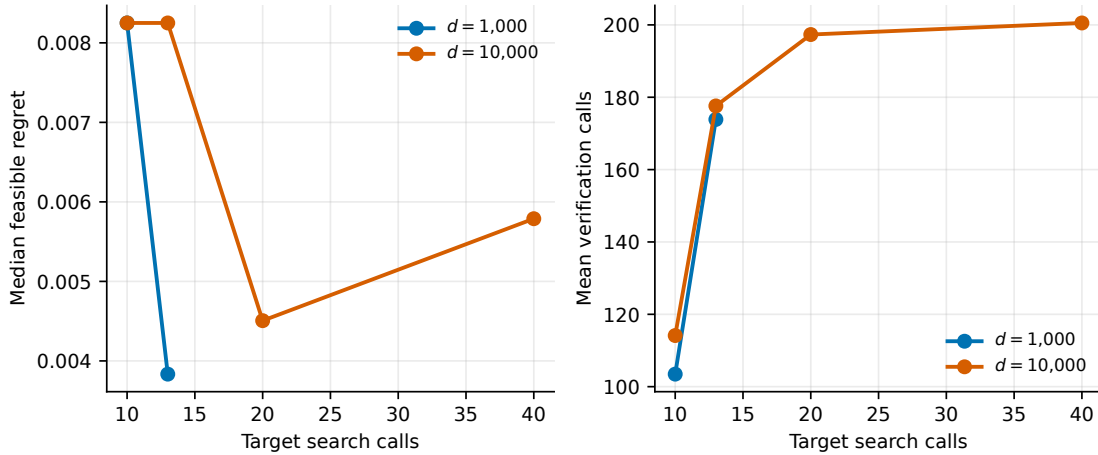


Figure 4: Dimension/budget frontier: aggregate median feasible regret (left) and mean verification calls (right). The $d = 10000$ result is an effective- dimension stress test under the registered structural family.

7.5 External Electricity Holdout

All three energy arms are independently certified in 20/20 seeds with no false certificate (Table 7). The frozen atlas at $N = 13$ beats equal-total-cost common Sobol at $N = 397$ in all 20 paired seeds. The median objective difference (atlas minus Sobol) is -0.05331 , with bootstrap 95% interval $[-0.05408, -0.05058]$ and one-sided exact sign $p = 9.54 \times 10^{-7}$.

The natural low-frequency grid changes the interpretation. The atlas wins 9/20 and loses 11/20; the median difference is 0.000945 , with 95% interval $[-0.00108, 0.00180]$ and one-sided sign $p = 0.748$. Thus the external result supports low-frequency structural compression and a large advantage over unstructured total-cost search. It does not establish that source learning outperforms an obvious domain-specific low-frequency control.

Table 7: Untouched Great Britain electricity-storage holdout and retained post-confirmatory fairness controls.

Arm	Certified	False	Median objective	Mean verify
Frozen atlas, $N = 13$	20/20	0	0.0976	80.0
Natural low-frequency grid, $N = 13$	20/20	0	0.0956	80.0
Common Sobol, $N = 397$	20/20	0	0.1488	80.0

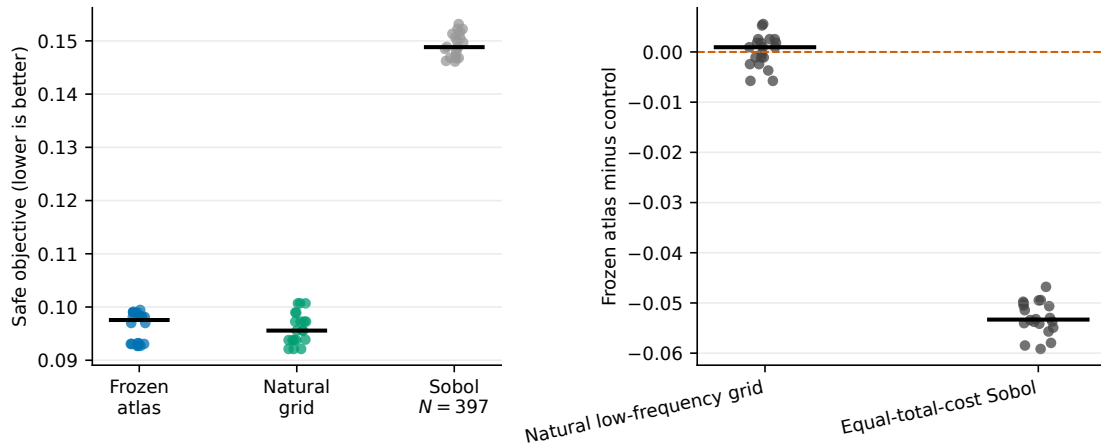


Figure 5: External energy paired differences. Negative values favor the frozen atlas. The left panel is a retained negative result; the right panel is the equal-total-cost unstructured control.

7.6 Heteroscedastic Decomposition Is a Calibration Diagnostic

The cumulative factor model sharply improves variance-shape recovery under a matched proposal, canonical SCBO backend, verifier, and 20-seed design. Mean log-variance RMSE falls from 1.158 to 0.219 in FactorShock, 0.577 to 0.349 in Inventory, and 0.501 to 0.254 in Queue. Shape correlation rises from zero under pooled variance to 0.919, 0.953, and 0.995. The Holm-adjusted Wilcoxon p -value is 1.14×10^{-10} for both log-RMSE and shape correlation.

Operationally, however, both models produce 59/60 feasible and certified recommendations, zero false certificates, and identical median paired regret. The HVD model also increases mean verification cost in Inventory and Queue. It therefore fails the preregistered promotion rule and remains an optional risk-calibration diagnostic.

Table 8: Pooled versus cumulative factor-HVD. Variance diagnostics are post-run only; they do not enter target selection.

Domain	Variance model	Feasible	Log-RMSE	Shape corr.	Mean verify
FactorShock	Pooled	20/20	1.158	0.000	159.2
FactorShock	Cumulative factor-HVD	20/20	0.219	0.919	158.4
Inventory	Pooled	20/20	0.577	0.000	182.4
Inventory	Cumulative factor-HVD	20/20	0.349	0.953	184.0
Queue	Pooled	19/20	0.501	0.000	190.4
Queue	Cumulative factor-HVD	19/20	0.254	0.995	197.6

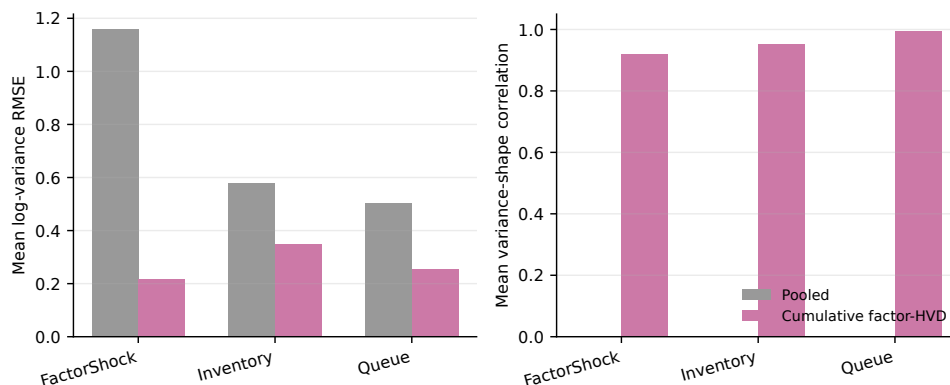


Figure 6: Cumulative factor-HVD recovers variance shape but does not improve feasible recovery or regret in the matched causal experiment.

8 Discussion

8.1 What the Experiments Identify

The central empirical finding is an attribution result. Under a target budget of 13 calls, the structural frontend determines whether the search observes a feasible basin; the online backend determines a smaller, conditional objective improvement after coverage. This distinction explains why several transfer baselines become strong when supplied with the same frozen atlas and why a powerful high-dimensional optimizer can fail under common Sobol. The proposed object is therefore not “SAASBO plus a warm start.” It is a reusable, source-trained experimental design whose value can be measured with the backend held fixed.

The component ablation also refines the structural story. A universal low- frequency library alone is too generic, and source templates alone are too brittle. Their union, followed by source-risk/objective ranking and maximin coverage, is robust across the three synthetic mechanisms. The external energy audit then reveals an important boundary: in a domain

with an obvious constant-profile control, a natural low-frequency grid is as good as the source-trained atlas. Source learning is valuable when it identifies a nontrivial transferable profile family, not merely because historical data exist.

8.2 Operational Meaning of the Cost Decomposition

The method trades reusable offline cost and explicit verification cost for a small target search budget. In a one-off target problem, $S = 384$ source calls must be counted in full. In a repeated deployment program, the same archive can support many target tasks and its per-target cost decreases, but this amortization is an operational decision rather than a statistical free pass. Likewise, independent verification often costs far more than the 13 search calls. That is not a defect hidden by the analysis: it is the price of a deployment statement that does not trust the optimizer’s posterior.

This decomposition is useful for managers choosing among protocols. If the decision is exploratory and no deployment certificate is required, proposal- only or atlas-plus-Sobol may be sufficient. If objective quality matters after coverage, canonical SAASBO adds value at higher computation cost. If an unsafe deployment has material consequences, verification should be budgeted as a separate final experiment rather than borrowed from search data.

8.3 Why the High-Dimensional Result Is Plausible

The raw search spaces are enormous, but the benchmark policies affect system behavior through coarse profile summaries. The source archive identifies which low-frequency profiles should be transferred, and the inverse map constructs those profiles at a new dimension. The relevant statistical dimension is therefore the geometry of $\hat{\eta}$, not the number of integer entries. Theorem 5.6 captures precisely this mechanism.

This mechanism is also falsifiable. If target safety depends on isolated high-frequency coordinates, if source and target safe supports are far apart, or if coordinate error consumes the safe depth, condition (10) fails. The finite-atlas no-free-lunch theorem then applies. We regard that explicit failure mode as preferable to an opaque claim of universal meta-learning.

8.4 Heteroscedasticity and Certification

The HVD diagnostic shows that variance structure can be statistically recoverable without changing the selected policy. This is unsurprising when the atlas already places a deep feasible point in the initial design and the terminal verifier can spend more samples. The decomposition may matter in applications where shared shocks dominate, safe depth is narrow,

or verification is more expensive; the present data do not establish such an optimization gain. We therefore keep Equation (12) and its formal properties as a secondary risk-modeling result, not as a reason to enlarge the paper’s main claim.

8.5 Limitations

Five limitations delimit external validity.

- (i) The three primary domains are synthetic and intentionally satisfy a related structural family. Their different operations interpretations do not replace independent real systems.
- (ii) The energy study contains one untouched market-year target. It supports low-frequency structural compression but not superiority to a natural structured control.
- (iii) The global margin Lipschitz constant and ideal-coordinate error are assumptions. The registered finite-library audit passes in all three domains, but a global certificate is unavailable.
- (iv) Verification controls its declared statistical event under fresh-sample independence. Distribution shift after deployment is outside that contract.
- (v) The source archive is fixed at two domains and 384 calls. Richer task collections could improve or destabilize the atlas; learning curves in the number and diversity of source tasks remain future work.

The failed traffic development gate is retained in the experiment registry and supports the same caution: not every domain exposes a transferable feasible profile at the chosen fidelity. We do not use that failed development study as positive evidence in this paper.

9 Conclusion

High-dimensional chance-constrained simulation optimization cannot be made sample efficient by acquisition design alone when a tiny target design misses the feasible region. We addressed that bottleneck with a source-learned, dimension-equivariant structural proposal atlas, a replaceable online backend, and an independent terminal verifier. The theoretical analysis first rules out unconditional finite-atlas coverage and then gives an explicit geometric condition under which coverage is independent of nominal policy dimension.

The experiments identify the atlas as the dominant source of feasible-basin coverage. At dimension 1,000 it succeeds across three backends and all 60 trials, while matched common-Sobol designs almost always fail. Canonical SAASBO improves objective quality after coverage, but is not essential to feasibility and is not a methodological contribution. The

dimension-10,000 and external energy studies support transfer of low-frequency structure, with the important qualification that source learning does not beat an obvious structured energy control. Cumulative heteroscedastic decomposition improves variance calibration but not optimization in the registered causal test.

The broader lesson is that historical simulations should be evaluated as an experimental-design resource: what policy geometry do they cover, what target assumption makes that geometry transferable, and how much independent evidence is required before deployment? Separating those questions from the optimizer produces stronger attribution, more honest cost accounting, and a method whose limits are as explicit as its gains.

A Additional Implementation Details

A.1 Synthetic Population Interfaces

All synthetic simulators return three Gaussian observations: two objective components and one constraint component. The scalar objective is the equal-weight average of the first two. The chance constraint uses the population mean and heteroscedastic standard deviation of the third as in (2). Each raw coordinate is integer valued between 0 and 100. The domain summaries are deterministic functions of the normalized policy and are available to the simulator; analytic target optima and feasibility labels are unavailable to the algorithm.

FactorShock uses a policy-level summary (u, q, s) and the cumulative exposure blocks

$$A = (u\text{-gap}, q\text{-gap}, \text{spread}), \quad N = (\text{oscillation}, \text{tail shock}).$$

Inventory partitions the policy into stock, reorder, and safety blocks; Queue partitions it into capacity, priority, and smoothing blocks. Each objective and safety pocket is smooth in its block means and dispersion, but the centers, loss weights, and variance mechanisms differ by domain. The post-run oracle enumerates the low-dimensional summary grid to compute the feasible optimum.

A.2 Canonical Backend and Discretization

Canonical SAASBO is implemented in BoTorch with a fully Bayesian single-task GP and sparse axis-aligned shrinkage prior. The model is refit after each of the three adaptive target evaluations. Proposed normalized policies are rounded to the target integer cube before simulation. The source archive is not appended as 384 pseudo-target observations; its primary role is to train the atlas. Transfer baselines use the archive according to their own model

contracts.

A.3 Terminal Verifier

The synthetic verifier freezes, deduplicates, and orders up to three policies: an objective challenger, the backend recommendation, and a safe-interior support candidate. Safety is evaluated with fresh Gaussian replications under a one-sided mean/scale tolerance rule and familywise error spending. Candidate budgets are 80, 128, and 128, and the verifier stops at the first certificate. When a challenger and incumbent are both admissible, each receives eight fresh objective replications and the challenger switches only when the one-sided upper bound on its paired cost difference is below zero. No verifier sample is returned to the surrogate, acquisition, shortlist construction, or source model.

B Machine-Checked Proof Map

The Lean 4 project contains 102 source files and builds against mathlib without `sorry`, `admit`, or project-defined `axiom`. Table 9 maps the claims used in the manuscript to their principal files. Lean proves the displayed implications; empirical assumptions such as source-target support shift remain scientific premises rather than axioms asserting that every domain satisfies them.

Table 9: Principal Lean proof files.

Claim	File under <code>proof/SCOLHKG/</code>
Finite-atlas no-free-lunch	<code>Real/ProposalNoFreeLunch.lean</code>
Aligned geometric atlas coverage	<code>Real/GeometricAtlasCoverage.lean</code>
Source-label noninterference	<code>Real/RiskAlignedRepresentation.lean</code>
Fail-closed monotone envelope	<code>Real/SourceMonotoneEnvelope.lean</code>
Shortlist safety and objective guard	<code>Real/MethodIndependent</code> <code>TerminalVerification.lean</code>
Exact binomial all-success certificate	<code>Measure/ExactBinomialCertificate.lean</code>
Composed paper interfaces and budget identity	<code>Real/PaperMainline.lean</code>

The final source-tree SHA-256 recorded by the proof receipt is:

```
536ca1d10e971f8001f4d38b
050508e2448ce31fb51ec8a9
abfdb0d6f343a710.
```

The receipt is regenerated only after a successful `lake build` and a zero-placeholder scan.

C Reproducibility and Data Contracts

Each result row records the task fingerprint, source-archive fingerprint, initial-design fingerprint, method identity, implementation commit, verifier signature, source/search/verification counts, and terminal truth reconstruction. The release audit requires every registered cell exactly once, identical problem fingerprints within paired comparisons, complete target-search traces, and agreement between stored and reconstructed truth.

Manuscript tables and figures are generated only from compact audited JSON and CSV artifacts. The renderer explicitly excludes runtime checkpoints, pickle files, model weights, and policy vectors, and writes SHA-256 hashes for every input and output. The experiment registry also retains named algorithm variants: periodic-capped SAAS is never relabeled as canonical SAASBO, and failed or incomplete development tracks are not silently merged into release results.

The code release includes the final method manifest, compact result records, paired-statistics artifact, convergence summaries, external energy data contract, proof receipt, and executable tests. The OPSD source data are cited by version and DOI (Open Power System Data, 2020). Runtime checkpoints and model weights are not required to reproduce the audited numerical tables and are excluded from the compact release.

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